

Apoorv Sharma

Software Engineer
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WORK EXPERIENCE

Amazon

Bellevue, WA, USA

Software Engineer

Dec 2024 – Present

- Led end-to-end delivery of a [policy-data pipeline](#) across three services, replacing a complex legacy model with a simplified schema; owned design, implementation, and a phased, [flag-gated rollout](#) that anchored a larger [platform migration](#).
- Root-caused a production thread-starvation outage in a [Kotlin](#) service to a nested run-blocking deadlocking the [coroutine dispatcher](#); confirmed via a load-tested revert and [drove stuck threads to zero](#).
- Built an automated API-throttling monitor ([CloudTrail](#) → [CloudWatch](#) → [scheduled Lambda](#)) that detects traffic nearing [AWS AppConfig](#) limits and uncovered a misconfiguration concentrating [7,000+ TPS](#) on the wrong endpoint.
- Designed a [cross-service signal](#) propagating intent metadata through a multi-system pipeline, fixing a gap where [~50% of intentionally delayed orders shipped early](#).

Clairvoyant Intelligence (formerly D&G Analytics, Inc.)

Rochester, NY, USA (Remote)

Software Engineer

May 2023 – Nov 2024

- Built virtualized infrastructure on [KubeVirt](#) and [CDI](#) over [Kubernetes](#) and [AWS](#), with an [Async-io](#) backend pipeline to deploy and manage Windows VMs for archive extraction jobs.
- Built a [Python](#)-based tool integrating 'Universal Extractor 2' and a custom Auto-It script for Windows GUI control, automating archive extractions for a [52% reduction](#) in manual processing time.
- Optimized the extraction process by focusing on file headers, which demonstrated an [83% increase in extraction efficiency](#).

Amdocs Development Center India LLP

Gurgaon, HR, India

Software Developer

Aug 2020 - Jul 2022

- Engineered a message handler for Telstra's OSS platform using [Spring Boot](#), bridging downstream [Apache Kafka](#) traffic with upstream [REST APIs](#) bi-directionally.
- Hardened [API security](#) by integrating Amdocs's core security layer, redirecting unauthorized requests before they reached [spring controllers](#) at the Kubernetes cluster layer.
- Led development of an automation framework using [Cucumber](#) and [TestNG](#), integrated with a [Spring Boot](#) application that [simulated Kafka transactions](#), and [slashing average testing duration by 70%](#).

SKILLS

Languages	Kotlin, Java, Python, Shell, SQL, JavaScript, Auto-It.
Tools	Docker, Kubernetes, GIT, Linux/Unix, Jenkins, CI/CD, Maven, Postman, Swagger, Github, JIRA, IntelliJ, VSCode.
Frameworks	Async-io, Tornado, NATS, Unittest, Spring-Boot, ReactJS, Apache Kafka, PyTorch, GraphQL, Cucumber, JUnit, TestNG.
Technologies	KubeVirt, CDI, Microservices, REST, OAuth, SAML, SOAP, Postgres, Design Patterns, Web services.
Methodologies	OOP, Agile, Data Structures and Algorithms, Distributed Systems, Concurrency & Coroutines, Deep Learning.
Cloud	AWS, OpenStack.

EDUCATION

University of Rochester

Rochester, NY, USA

Master of Science, Computer Science

Aug 2022 - Dec 2023

Thapar Institute of Engineering and Technology

Patiala, PB, India

Bachelor of Engineering, Computer Engineering

Jul 2016 - Jul 2020

ACADEMIC PROJECTS

Pose estimation for fitness routines

Oct 2023 - Dec 2023

- Implemented and analyzed the performance of two state-of-the-art pose estimation methods – '[Mediapipe](#)' and '[YOLO-NAS](#)'.
- Analyzed the methods on qualitative metrics like - 'low-light', 'occlusion' and 'distance' and quantitative metrics like [PCK](#) and [MSE](#).
- Concluded that [Mediapipe](#) is better for lightweight implementation, while [YOLO-NAS](#) is better when accuracy is a crucial aspect.

Sequential Modelling

Oct 2022 - Dec 2022

- Engineered a system implementing two [Hidden Markov Models \(HMMs\)](#) with configurable hidden states.
- Trained the models using [Expectation-Maximization algorithm](#) and [Sum-Product algorithm](#), aimed for improved natural language classification, [solely utilizing the NumPy](#) library.